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Computer Vision Based Mouse With Color Detection Technique

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***Abstract*—In this research paper, we have used computer vision technology for the development of interaction between computers and humans. This technology can act as an interface between computers and humans. Nowadays, everyone uses mouse and keyboards to interact with computers. This traditional approach of interaction is costly as well as not a user-friendly approach. In this paper, we propose a computer vision-based mouse with a color detection technique that uses a web camera of our computer to detect the color pointer and accordingly it perform the various mouse functions on the screen. This developed approach of performing different mouse functions can reduce the hardware cost of the computer and make it user-friendly for everyone.**

Keywords—Computer Vision, Color Recognition, Web Camera, Marker Motion Tracking, Cursor Control.

I. INTRODUCTION

Nowadays, everyone uses the computer to perform different tasks in daily life. Everyone must need some devices which can be easily used as an interface between humans and computers. The devices like mouse and keyboards are used at the input side to perform various tasks in the computers. These input devices increase the cost of the computers and also require more space than the display of the computer. Some people are not even comfortable with the touchpad of the laptop. Some people are also there who have a specific type of disability in their hands or fingers. Those individuals who are affected with some disability are unable to use input devices like mouse and keyboards to perform various tasks on the computer. We will use the new technology of interactions instead of using physical mouse and keyboards.

Many researchers have analyzed their research in the disciplines of human-computer interaction (HCI), virtual environment and in robotics, have tried utilising gestures, video devices and finger tips to operate the mouse's functionalities. But, all the researchers have suggested different techniques to control the mouse functionalities. In 2002, Erdem *et al.* [1] have analyzed controlling the movement of mouse using fingertip tracking. To perform one of the clicking action of the mouse, the user must merely place his or her finger on one of those areas or regions corresponding to mouse buttons [1][6]. Another researcher, Chu-Feng Lien [4] proposed a different method. He simply used his fingertips to manage the movement of the mouse cursor and the clicking action of mouse. His technique of clicking was developed on density of the image, and it asks the user to keep the mouse pointer on the target area for a short moment. Also, Hojoon Park [2], have used different method in which he used the index finger for the movement of the mouse cursor and the inclination between the index finger and thumb is utilised for different clicking events of the mouse. Also, Kamran Niyazi *et al.* [3] developed a new approach in which they used index finger with yellow colour tape to keep the movement of mouse cursor under control and used the distance between the two coloured tapes in the fingers to perform the clicking events. C. Dhule *et al.* [5] have analyzed another approach in which motion of hand and fingers determine the change in pixels values of RBG colors from a real time video is used to manage the motion of the mouse. In 2004, Paul Robertson *et al.* [6] has described a different method to click in which they have implemented the thumb's motion (from a 'thumbs-up' posture to a fist) to define a clicking activity and a special hand-sign is used for the movement of mouse cursor. D. Firmanda *et al.* [7] have analyzed and defined a different approach in which they have used object tracking methods based colors by sensor camera to control the movement of cursor. Y. Sato *et al.* [8] have defined the quick and reliable method for detecting the locations of the centres and fingers of both the hands. P. Anandan [9] in his paper provides a hierarchical computational framework for determining dense displacement fields from a pair of pictures, as well as an algorithm based on it by which visual motion may be accurately measured using digital picture sequences. Andrew D. Wilson *et al.* [10] in their paper described the optical flow method which is used to analyze the motion of the hand as well as a capacitive touch sensor to activate and deactivate the interactions. R.Kjeldsen *et al.* [11] have analyzed the implementation of a workable system for controlling a window-based user interface by visually recognising hand movements. Andrew Wilson *et al.* [12] have proposed a method of stereo vision in the paper which allows for rapid and reliable in-depth sensing of the user and serves as a helpful resource for developing future perceptual user interfaces by which traditional mouse and keyboard interfaces can be eliminated. Antonis A.Argyros *et al.* [13] have created a vision-based interface for using 2D and 3D hand motions to operate a computer mouse. Marco Porta [14] have researched and presented a broad insight into the issue of image processing, vision-based interfaces (VBIs) and head tracking, face/facial expression recognition, eye tracking, and gesture recognition are the four primary domains of that technology. By analyzing the methods utilized to interface them. Antonis A.Argyros *et al.* [15] have presented the scheme consists of a set of algorithms for modelling and detecting skin-colored objects in picture sequences, as well as their temporal relationship. Gang-Zeng Mao *et al.* [16] have observed that combining a modified object detection approach developed by Viola and Jones [19] with a skin-color detection method to execute hand detection and

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tracking against a complicated background image have been found to be effective. Mathias Kolsch *et al.* [17] have described in the research that a collection of general gesture-based interactive methods and have analyzed their properties in terms of task compatibility and the necessary computer vision algorithms to recognise them. These methods, when combined with other algorithms, achieve usability and involvement even when both the camera and the target of interest are moving, as in mobile and wearable computing. Zhigeng Pan *et al.* [18] have analysed that multiple gestures hand tracking technique achieves real-time results in a continual manner despite the fact that the background is congested and this technique satisfies the needs of vision-based human-computer interactions with a frontal view in terms of speed and accuracy. Paul Viola *et al.* [19] have developed a visual object recognition framework that can analyse images quickly while achieving high detection rates. Ruchi Manish Gurav *et al.* [21] have defined a system that recognises gestures in real time using finger tracking and also used contour detection technique to differentiate between two different gestures. R. Cutler *et al.* [22] have analyzed a view-based motion detection system that detects movements using such a rule-based method based on motion blobs properties such as random movement and size. J. J. Corso *et al.* [23] have developed a 4D touchpad solution for human-machine interfaces that enables the ability for strong, unconstrained activity which is formed on the VICs framework's 3D-2D Projection-based method. Rick Kjeldsen *et al.* [24] have developed the methodology and framework for a computer vision system that allows for changing interface reconfiguration, altering the shape and position of interactions on the plane. J.M. Rehg *et al.* [25] have developed DigitEyes method which is a non-invasive vision based hand tracking technique and this method has proven the tracking capability at rates up to 10 Hz using line and point characteristics retrieved from grayscale pictures of bare and unmarked hands. David G.Lowe [26] have proposed a real-time motion tracking of three dimensional objects which enables the resolution of matching and measurement problems that occur during motion tracking in a specific order. M. Yamamoto *et al.* [27] developed a robotic arm model that defines the human body motion. Nasser H. Dardas *et al.* [28] have analyzed and provided a real-time technology for using hand gestures to connect with an application or video game. Qing Chen *et al.* [29] have proposed a two-level solution to the real-time vision-based hand movement interpretation problem. The posture recognition with Haar-like features and the AdaBoost learning algorithm are performed at the lower level of the method while the upper level uses a context-free grammar-based syntactic analysis to achieve linguistic hand gesture recognition. M. Kolsch *et al.* [30] have developed an approach based on frequency analysis for estimating class separability in real time without the requirement for any training.

Our research was motivated from papers of Andrew Wilson *et al.* [12] and Gang-Zeng Mao *et al.* [16]. They have developed a vision technique for building future perceptual user interfaces by which traditional mouse and keyboard interfaces can be eliminated [12]. They have used analyzed the tracking and detection in complex background. Our project used the marker motion tracking along with the color recognition feature.

In our project, we will design a computer vision-based mouse with a color detection technique using the web camera installed on the computers. We will use the OpenCV-Python to perform the computer vision operation in our project.

Computer vision is the most significant approach [20]. Computer Vision, in the easiest form, is the automation of such a visual system, so that computers or machines, in general, can obtain high-level understandings of the understandings from Digital Images and Videos [20]. The purpose of computer vision is to know the content of the pictures. It extracts the description from the pictures, which may be an object, a text description, and three-dimension model, and so on.

This computer vision technology can be used in various types of applications like in machine vision, in object detection, in activity recognition, in human-computer interaction(HCI) devices, and autonomous navigations [20].

Following pipeline in “Fig. 1” is used for computer vision application :

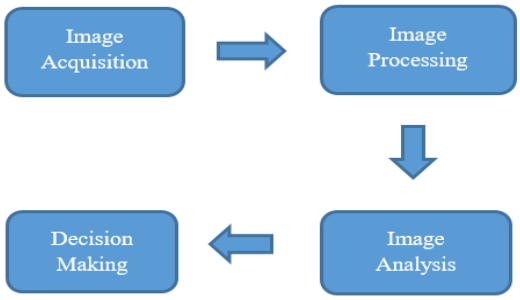


Fig. 1. Pipeline for computer vision applications

The above pipeline illustrates the process involved in computer vision application. Any computer vision application starts with image acquisition. Image Acquisition is the digital representation of the visual characteristics of the physical world. After that, the acquired image is processed to the image processing. At this stage, the signals in the acquired images are filtered to remove the noise or any irrelevant frequencies. Then these processed images are analyzed to extract useful information at the image analysis stage, which involves pattern identification, color recognition, object recognition, feature extraction, motion tracking, and image segmentation. Finally, at the decision-making stage, the high dimensional data obtained from all the previous steps are used to produce meaningful numerical information that leads to decision making.

This computer vision algorithm helped us to overcome problems like object recognition and classification, pose estimation, optical character recognition, facial recognition, motion tracking, scene reconstruction, and image restoration, which we are facing earlier before the use of this computer vision[20].

In this project, we have used an OpenCV tool for the use of computer vision [20]. OpenCV stands for Open Source Computer Vision [20]. It is a library developed in c++ and composed of functions that enable real-time computer vision. OpenCV library is supported in popular programming

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languages like python, java, matlab and octave[20]. There are some reasons for using only OpenCV tool which are:

(i) Open CV is available for free of cost.

(ii) Open CV library is written in C/C++, so it is quite fast.

(iii) Less RAM

(iv) a usage, maybe of 60-70 MB.

(v) Computer Vision is portable as OpenCV and can run on any device that can run on C.

In this project, we use a python programming language with the OpenCV library.

II. METHODOLOGY

The pipeline shown in "Fig. 2" describes the process involved in our project.

(i) Image Acquisition

Image Acquisition is the digital representation of the visual characteristics of the physical world. We use image sensors to detect and to capture the required information to make an image. It will detect the colored markers present on our palm that needs to be tracked. This process is responsible to capture the real-time videos and to read the frames from the videos [20]. After obtaining the frames from the videos, now, we can apply the image processing technique to the individual frame. These frames are processed to the next step.

(ii) Image Preparation

Image preparation is used to prepare the raw images. This raw image is smoothened by using gaussian smoothing. Smoothing is very useful for removing noise from the images. It removes the high frequencies of contents like noise and edges from the images. It is also effective in reducing pixelation of an image. Smoothing replaces the pixel values with the 'average' of the values of the pixels in its neighborhood. The 'average' is calculated in different techniques, which leads to different outcomes in smoothing. There are various types of smoothings are available. But, we have used the gaussian smoothing, which smoothens the image using a gaussian filter, which is a weighted average of the intensities in the pixel neighborhood. Now, define a Region of Interest (ROI) and track the motion of the colored marker in that region. Now, we will form a rectangle and a grid on the actual frame to indicate this region of interest.

(iii) Color Recognition

After the image preparation process, now we will recognize the colored object in the image. In this project, we will track the object-based on their specific color. In OpenCV, the images are by default in the BGR colorspace format [20]. This color recognition is easier and better to identify in the HSV colorspace. So, we will convert the image from BGR to HSV colorspace. We will separate the objects of a specific color or a color range from the image. Finally, we will do a thresholding operation based on the marker color to identify the objects of the specific color range.

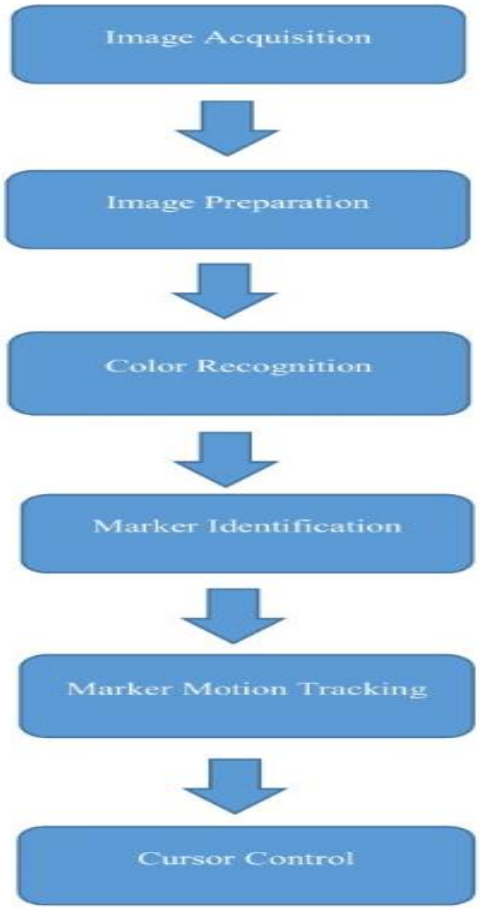


Fig. 2. Pipeline used in CV Based Mouse

(iv) Marker Identification

Marker identification is used to identify the different types of colored markers from all the objects recognized in the images. We have converted the original image into a binary image and we will find the contours of the colored objects, from the binary image. After that, we will find the contours with the maximum area, as the contour of interest and identify it as the object.

(v) Marker Motion Tracking

Once we identify the markers in each frame then we will track the motion of the markers in different frames by locating the centroid of the marker relative to the grid in the Region of Interest.

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(vi) Cursor Control

Finally, we will control the movement of the cursor on our computer screen based on the marker motion. We will also perform different click actions like left-clicking, right-click and double-clicking of left mouse button based on the color of the markers [2].

Earlier we have seen the pipeline used in our project. But, now, we will discuss how our project works along with a brief description of the block diagram.

Following are the steps used in the block diagram described below in "Fig. 3".

a) Real-Time Video Capture

The web-camera of our computer is used to capture the real-time video as an input [2]. Video is a collection of several visual images. Videos by the camera varied in terms of frame rate, aspect ratio, display resolution, and refresh rate. The frame rate is the frequency at which consecutive pictures called frames seems on the display and is expressed in frames per second (fps).

b) Range of the colors of the markers

In our project, we have used four different colored markers to perform different mouse functions on the computer screen. We have defined the range of the colored markers in the HSV colorspace. In HSV colorspace, the range of the colors varies from 0 to 255 where lower value 0 shows the full black color and the higher value 255 shows the full white color.

c) Flipping of frame

In our project, we are using the web-camera of our computer which, by default, shows the mirror image and so the marker motion will also be flipped.This will make the cursor move in the opposite direction. It means if we move our marker from left to right then on the screen marker will move in the opposite direction from right to left. So, to overcome this problem, we will flip the frame of our camera and make the interface more intuitive.

d) Image Smoothing and Region of Interest

Image smoothing is useful for removing the noise from the images. It removes the high-frequency content like noise and edges from the images. Here, we will use the Gaussian smoothing technique that smoothens the image by using a gaussian filter, which is a weighted average of the intensities in the pixel neighborhood. After smoothing the image using Gaussian smoothing, we will define a Region of Interest and track the motion of the marker only in that region.

e) Conversion of the colorspace of the masked image

In OpenCV, the images are by default in the BGR colorspace format. The color recognition is easier and better to identify in the HSV colorspace. So, we will convert the colorspace of the masked image from BGR to HSV.

f) Image Thresholding

Image thresholding is a technique used to convert a colored image into a binary image. It is primarily used in image segmentation, to segment an image into different areas based on specific criteria and has only those areas visible in the image for further analysis. Every pixel of an image has some intensity associated with it. The colored markers are bounded with the lower and the upper boundary of the HSV values, defined earlier, and this range helps to convert the HSV image into a binary image.

g) Contour for the colored markers

Contours are defined as the curves joining all the continuous points having the same color or intensity. The contours are useful for shape analysis and object detection & recognition. Here, we will calculate the centroid of the contour and use it to track the motion of the marker on the screen.

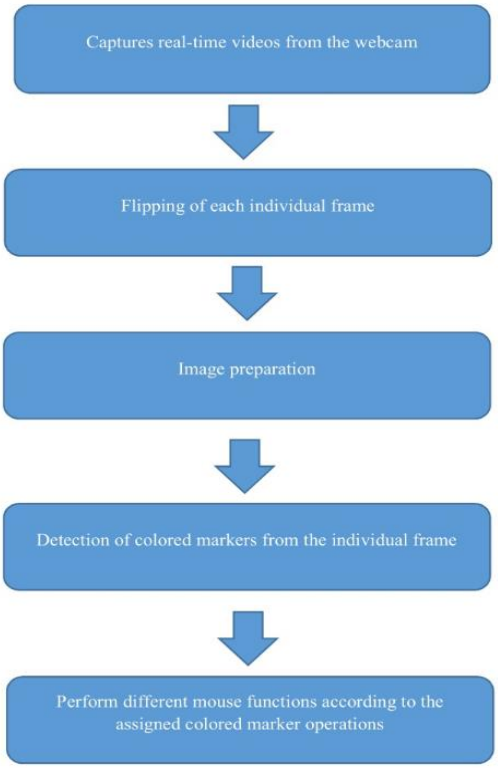


Fig. 3. Basic Block Diagram of the CV Based Mouse

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h) Performing mouse functions

Finally, after recognizing the color of the marker, different mouse functions are performed. We are using the yellow color marker to move the cursor on our screen [21]. To perform different clicking functions like left-clicking, right-click and double click of the left mouse button, different color markers are assigned and accordingly the mouse functions are performed [2]. Whenever the green-colored marker is detected then the Left Click function is performed. Whenever blue color marker is detected then the Right Click function is performed. To perform the Double Click function of the left mouse button, red color is used.

III. ALGORITHM

Following are the steps involved in the algorithm of our project:

(i) Captures the real-time video by the web-camera.
(ii) Apply image processing technique on the individual frame obtained from the live videos.
(iii) Assign a range for the colors of the markers used in our project.
(iv) Flip the frame about the vertical axis to avoid the problem of mirror effect in the image.
(v) Using Guassian Smoothing to apply the guassian filter to the image.
(vi) Define a Region of Interest to track the motion of the marker only in that region.
(vii) Conversion of the masked image into HSV colorspace.
(viii) Recognize the colors of the markers from the image.
(ix) Determine the position of the colored marker and calculate the centroid of the same.
(x) Performing different mouse functionalities with the color detected on the computer screen.

IV. ADVANTAGES AND LIMITATIONS

In our project, we are using different color markers to perform different mouse functions, which is a contactless human-computer interaction. Using these color markers can reduce the cost of our system by avoiding the use of physical devices as input. It can also be helpful for the persons who are not comfortable in using the touchpads.

Our project is dependent on the color detected on the screen. If there are multiple colors are available in the background then multiple colors will be detected on the screen which will cause the detection problem in our project. So, to avoid all these problems, we should make sure that there is no interference with any background objects, along with our color markers. We can also use the best thresholding method with the right threshold values.

V. RESULTS

Hence, we have successfully implemented computer vision technology in our project by using the web-camera of our computer, so that interaction with computers becomes easier. In our research paper, we have performed different mouse functionalities with the color marker detected on our screen. We have assigned four different colors to perform four different functions of our computer vision-based mouse. For moving the cursor on our screen, we have used a yellow color in “Fig. 4”. To perform the Left Click function of our mouse, we have used green color in “Fig. 5”. To perform the Right Click function, we have used blue color in “Fig. 6”. To perform Double Click of Left mouse button, we have used red color in “Fig. 7”. While using color markers, we should make sure that the background must be clear, so that our CV based mouse will work properly.

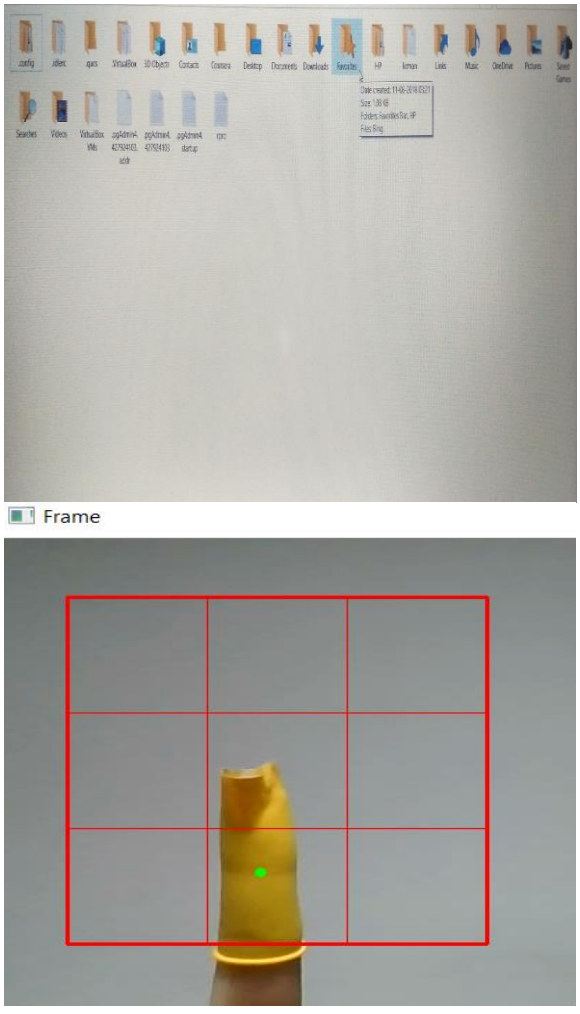


Fig. 4. Movement of mouse cursor using yellow color marker.

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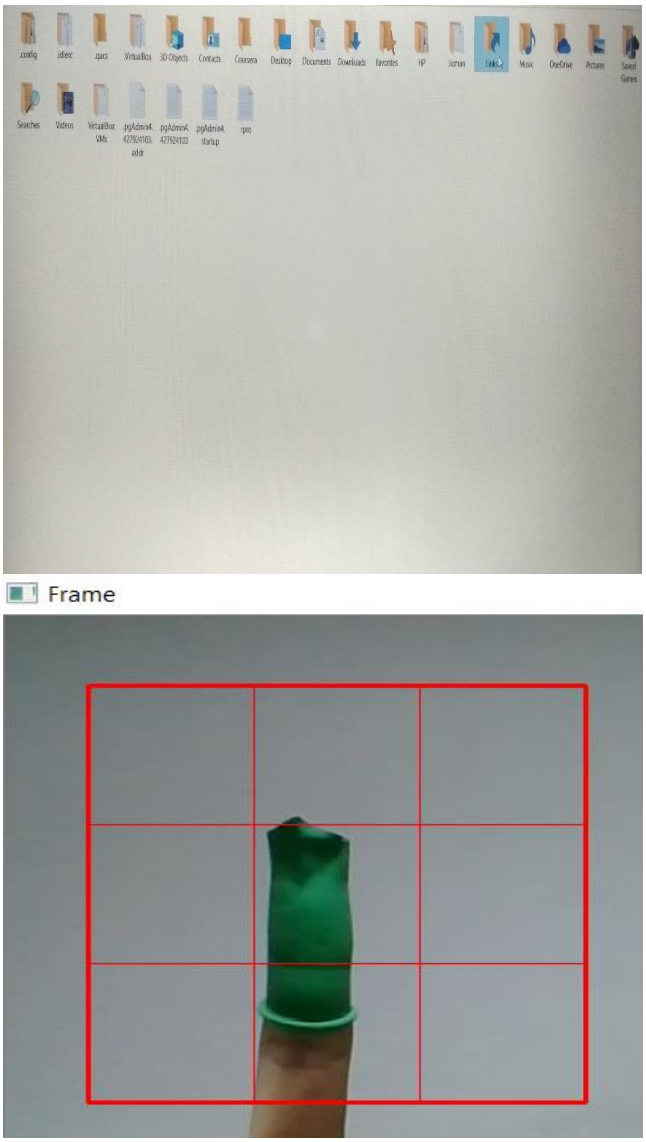
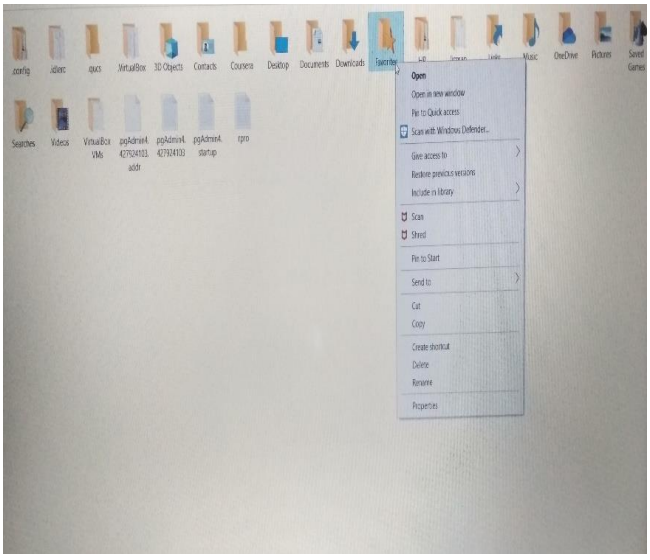


Fig. 5. Functioning of Left Click using green color marker.



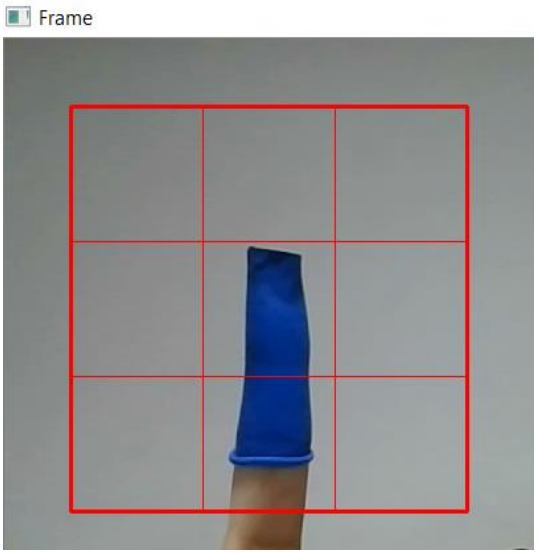


Fig. 6. Functioning of Right Click using blue color marker.

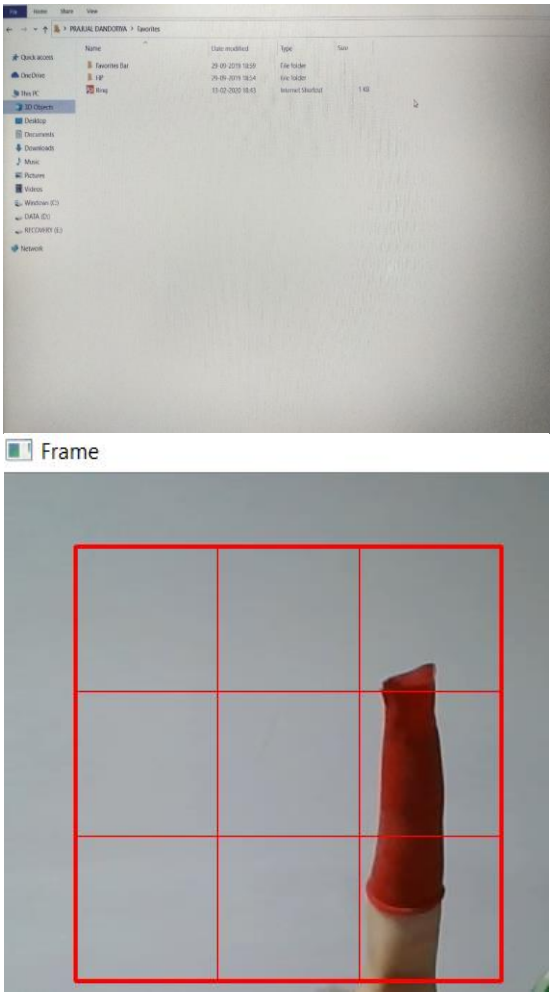


Fig. 7. Functioning of Double Clicking of left mouse button using red color marker.

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VI. FUTURE POSSIBILITIES

In the future, we can add more functionalities to our CV based mouse. We can use good thresholding methods to avoid background interference. We can also develop a new algorithm so that we can perform different mouse functions and other functionalities on our computer by using our voice. Our computer can be easily operated by using voice assistant as an input to the computer. This voice assistant on our computer will help many people who have a specific type of disability in their body. This type of modification in our project will be very useful for everyone in the future.

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